Quantifying Attribution of Annotator Polarization to Sociodemographic Groups

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: Polarization, annotation, annotators, socio-demographic, metric

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields such as content moderation, toxicity detection and hate speech detection. These tasks are especially essential for training systems that protect vulnerable and minority groups in online spaces, where they are often targeted [1, 2]. However, annotations for such tasks are subjective and usually based on majority-group annotators. This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems.

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree in vain. An example of such cases would be racist comments that are presented with coded language (usually referred to as "dog-whistles" [3]), which are

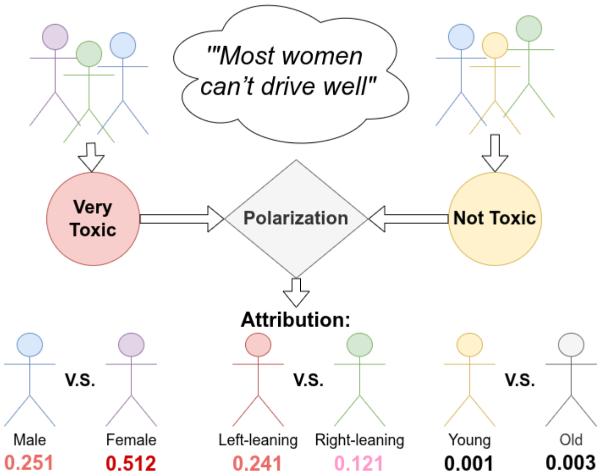


Fig. 1 The Apunim framework attributes annotator polarization to certain characteristics.

not picked up by most annotators, but only by ones belonging in the targeted minority. Inversely, majority-group annotators may bias their own annotations against the minority-group annotators. [4] for instance show that toxicity models disproportionally marked African American speech as "toxic".

Inter-annotator agreement is often used to detect such cases [5]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [6, 7]—specifically whether two or more annotation clusters are present (Figure 2).

While we can detect polarization [6, 7], there is little work on how to detect whether it is actually caused by marginalized groups. [7] propose a mechanism which only works in the extreme case where overall polarization is zero; in this work we demonstrate that this case is not frequent in real-world data. We instead propose the *Apunim ("Aposteriori Unimodality") metric*, which attributes polarization to individual annotator sociodemographic groups. Our contributions are:

1. We introduce a new metric which measures how much polarization can be attributed specific sociodemographic groups.
2. We discover issues not yet acknowledged in literature, such as the presence of inherent polarization and conflicting polarization directions. Our proposed metric is designed to avoid these issues.
3. We apply our metric to four datasets; two with human comments and annotations [8, 9] as well as two generated and annotated by Large Language Model (LLM) agents [10].

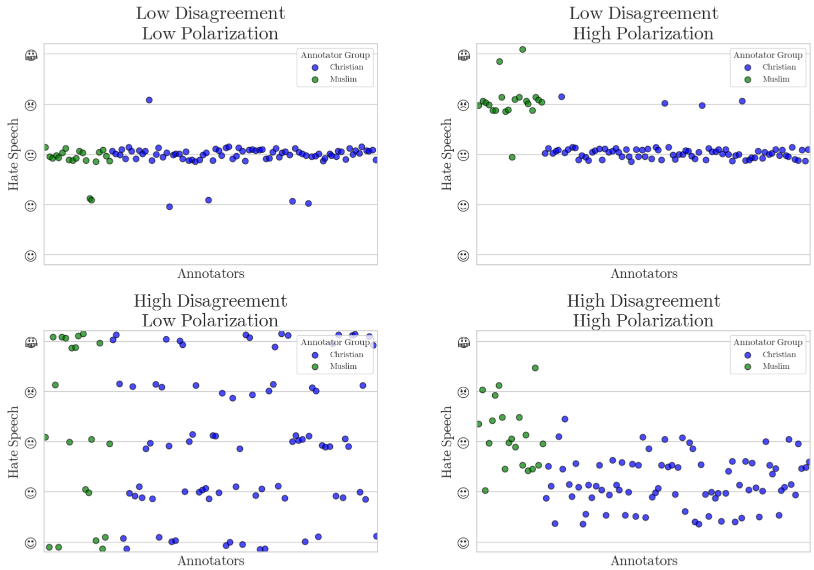


Fig. 2 Disagreement is similar to variance; thus it overlooks cases where annotators may be "split down the middle". Polarization on the other hand, is able to identify multiple clusters even if they are close together.

2 Methodology

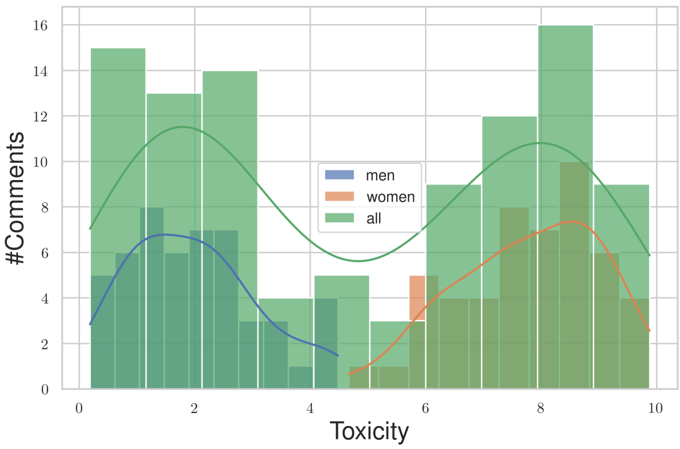
2.1 Problem Formulation

Sociodemographic Backgrounds

Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to isolate individual groups within a SocioDemographic Background (SDB). Let Θ be one of multiple "dimensions" the set of all annotator SDB groups for a single dimension (e.g. "male", "female" and "non-binary" if we are investigating annotator gender).

Annotations

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated datapoints. These datapoints could be of any modality, such as comments in a discussion, images, videos or survey questions. We assume that how polarizing a datapoint is depends on two variables: *inherent ("apriori") polarization*, and polarization caused by *different annotator SDBs ("aposteriori" polarization)*. Assuming that each datapoint c is assigned multiple annotators, we can define the annotation multi-set $A(c) = \{(a, \theta)\}$, where a



$\mathbf{nDFU_{men}} = 0.375$
$\mathbf{nDFU_{women}} = 0.100$
$\mathbf{nDFU_{all}} = 0.625$

Fig. 3 Example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

is a single annotation. As an example, the annotations for a comment in a sentiment analysis task would be formulated as:

$$A(\textit{“Could be better, could be worse”}) = \{(+, F), (-, M), (-, F), \ldots\} \quad (1)$$

Polarization

Each set of annotations features a certain degree of *polarization*. Unpolarized annotations are usually unimodal; the annotators disagree on the details (which would be shown roughly as a bell curve around the median annotation), not fundamentally (which would likely be shown as a multimodal distribution). [7] create a polarization metric, the “normalized Distance From Unimodality (nDFU)”, which measures whether an annotation set shows no polarization (unimodal distribution—$nDFU \approx 0$), up to complete polarization (multimodal distribution—$nDFU \approx 1$). Thus, we need a metric that measures how much the nDFU of a polarizing data point can be partially explained by θ.

2.2 Intuition

Datapoint polarization

Intuitively, θ partially explains the polarization of a datapoint c when the annotations grouped by θ are more polarized compared to the full set of annotations. Figure 3 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators[1]. The annotations are generally polarized

[1]The use of only two predominant genders is made only for the purposes of demonstration.

($nDFU_{all} = 0.625$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.1$), since most agree the data-point is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.3725$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

Previous work [7] stopped on a datapoint-level analysis. In practice however, this kind of analysis is inherently noisy due to a lack of information (i.e., annotations) relating to each individual datapoint. In order to get meaningful information, we need to consider information *between* datapoints, as the number of datapoints is usually several orders of magnitude larger than the number of annotations per datapoint (§3). In other words, we need to run our analysis on the dataset level.

Dataset polarization

Given this observation, we would be tempted to aggregate all annotations in the dataset and compare the polarization of each θ with the full set of annotations $A(c)$. In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each data point may be annotated by only a handful of annotators (5 is a common number in many publications), leading to a best-case even split between groups (for 5 annotators, a best case scenario would be a 3-2 split). Since nDFU is at its core a histogram estimation, 3 annotations may not be enough to acquire a reliable histogram (this is discussed in depth in §5.

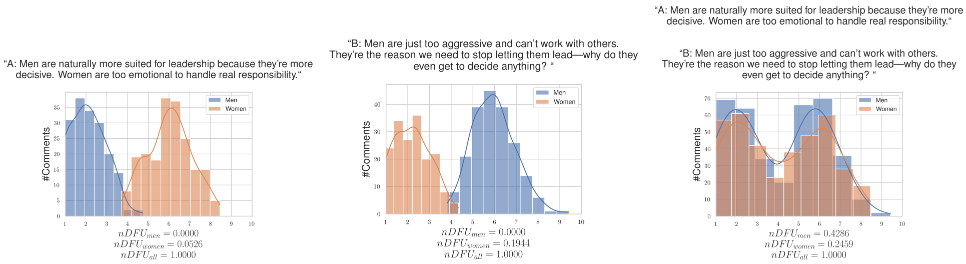


Fig. 4 Hypothetical example of a polarizing discussion with two comments, for which the annotators disagree on which is the toxic one. If we aggregate the two comments, the polarization scores for both men and women significantly rise, obscuring whether the exhibited polarization can be partially attributed to gender.

However, arbitrarily combining annotations would not work well in practice. Figure 4 illustrates a hypothetical discussion with two comments, both of which are toxic, but where male and female annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This example demonstrates that polarization has a *direction*, which is not captured by the (symmetric) $nDFU$ metric.

2.3 Aposteriori polarization

How polarizing a datapoint $c \in d$ is for annotators of group θ can be directly computed as:

$$pol(c, \theta) = nDFU(P(A(c), \theta) \tag{2}$$

where $P(A(c), \theta_i) = \{a(c; \theta) \in A(c) | \theta = \theta_i\}$ is the partition of A for a datapoint c, for which its annotators belong to the SDB group θ_i.

In §1 we mentioned that annotator polarization can be caused either by the annotators SDB, or other factors. The latter can be considered "apriori" polarization, which is driven by the inherent "controversy" of the data point. Previous work [7] considered the case where this apriori polarization was zero; this is not the case in many datasets, as we demonstrate in §3.

In a scenario in which each datapoint was annotated with a very large amount of annotations, we could use bootstrapping to compare the apriori and aposteriori polarization for each comment. This however is an unrealistic assumption [11]. Thus, instead of selecting subsets of annotations, we randomly partition the annotations in $|\Theta|$ groups, with matching group sizes,[2] which effectively "breaks up" the effect of a single isolated dimension on the polarization of the sample. The randomly partitioned annotations are then sampled t times. Formally, the average apriori polarization for a single datapoint will be given by:

$$\frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{3}$$

where $\tilde{P}_i$ is the random partition operator with matching group sizes.

While the low number of annotators is an important limitation when acquiring the observed polarization [11], this is decisively not the case with the number of data points themselves which can range up to hundreds of thousands. Thus, we have the luxury of filtering data points before computing a dataset-wide metric. Since our metric attributes polarization, unpolarizing datapoints are useless. The same applies for datapoints which are composed of only a single annotator group. We define the subset S_d of dataset d such that:

$$S_d = \{c \in d \mid nDFU(c) > \alpha \textit{ and } c \textit{ annotated by at least two groups}\} \tag{4}$$

where α is a small positive number.

By obtaining the pol-statistics for all data-points in S_d for SDB θ, we can get the mean observed polarization for the dataset:

$$pol_{obs.}(d, \theta) = \frac{1}{|d|} \sum_{c \in S_d} pol(c, \theta) \tag{5}$$

[2] E.g., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20.

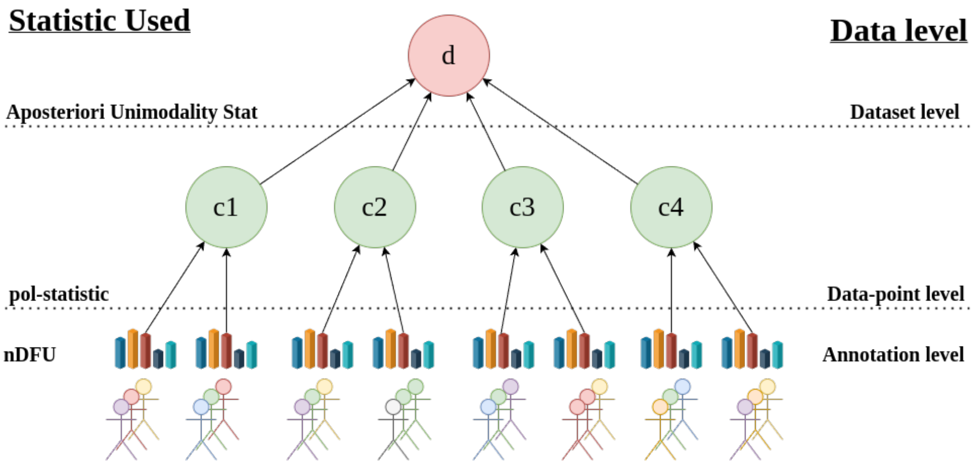


Fig. 5 An overview of the Aposteriori Unimodality Test. We gather statistical information from the annotations (different SDBs are denoted by different colors) via the nDFU measure, which is aggregated on the data-point-level by the pol-statistic. The Aposteriori Unimodality Test is applied on the discussion level, operating on the pol-statistics of the individual data-points.

The estimated average apriori polarization

$$pol_{apriori}(d) = \frac{1}{|d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{6}$$

We can then define the *"Aposteriori Unimodality" (Apunim)* Statistic as:

$$apunim(d, \theta) = \frac{pol_{obs.}(d, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)} \tag{7}$$

Since $nDFU \rightarrow [0, 1] \Rightarrow apunim \rightarrow [-1, 1]$.

If $apunim(d, \theta) \approx 0$, the exhibited polarization among annotators of group θ does not meaningfully differ from all other groups.

If $apunim(d, \theta) > 0$, then the polarization observed in the annotation task rises when this group is added to the annotation pool (equivalently, intragroup polarization is less than intergroup polarization). Example: If we observe $apunim(d, afr.am) = 0.3$, $apunim(d, white) = 0.001$ in a hate-speech detection task, it is likely that African American annotators fundamentally disagree with white annotators on what constitutes racism in the dataset d.

Conversly, if $apunim(d, \theta) < 0$, then the overall polarization would actually drop had we removed this group from the annotation task. Mathematically, this occurs when *apriori polarization is higher than the observed polarization* (intragroup polarization is higher than integroup polarization). In other words, annotators of that group may be very polarized between themselves. Example: If we observe $apunim(d, latino) = -0.3$, $apunim(d, white) = 0.001$ in a hate-speech detection task, it is likely that Latino

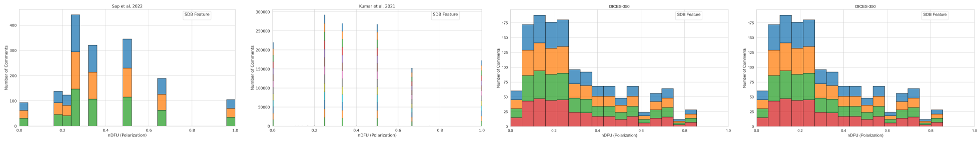


Fig. 6 Histogram of dataset polarization for the human datasets considered in this paper per SDB feature.

annotators fundamentally disagree with *each other* (compared to African American or white annotators) on which texts in the dataset d constitutes racism. This may indicate that other dimensions (confounding variables) are exerting heavy influence on the sample.

The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 5. There is a theoretical statistical issue for subtracting statistics derived from a subset and its superset, which is discussed in Appendix B.

2.4 p-value estimation

By obtaining the pol-statistics for all data-points in a dataset d we can apply any statistical means test with the null hypothesis:

$$H_0 : \forall\theta \in \Theta,\ apunim(d,\ \theta) \leq 0 \tag{8}$$

versus the (multiple) alternative hypotheses:

$$H_i : apunim(d,\ \theta_i) > 0 \tag{9}$$

Our test assumes that (1) there exist enough data-points to reliably run the means-test, and (2) enough annotations in each data-point and SDB group to estimate the *pol-statistics*. We discuss the former assumption in Appendix 5. The latter can be safely assumed for most datasets, because they feature enough data-points to reliably assume normal residuals ($n \gg 30$—see §3), which is why we use the Student-T test [12] .

Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms method [13]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [14]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

3 Datasets

We use the datasets provided by [9] and [8]. They both feature various online comments, each annotated by a number of human annotators (5 and 4-6 annotators per comment respectively). The Kumar [9] dataset stands out for featuring more than 100,000 comments, while also including 10 different dimensions for SDBs and annotator experiences. We also use the DICES-350 and DICES-990 datasets [15], which

Table 1 Brief overview of the datasets used in this study. We are interested in the number of comments (#Comm.), annotators per comment (#Ann.) and number of SDB dimensions (#Dims.).

Name	# Comm.	# Ann.	# Dims
Kumar [9]	106,035	5-6	10
Sap [8]	626	5-6	4
DICES-350 [15]	72,103	60-70	2
DICES-990 [15]	43,050	123	2

are concerned with LLM response safety, considering only labels relating to racism (`Q3_bias_overall`). They feature less annotated comments, but more than an order of magnitude more annotators per comment, making them an invaluable resource for our study. While the datasets technically target different annotation criteria (racism, toxicity and LLM safety respectively), these criteria are closely related to each other.

Brief dataset characteristics are given in Table 1. There is a reasonable amount of polarization in most annotations for all datasets as demonstrated by Figure 6. For in-depth information and discussion around our decisions for discretization, refer to Appendix C.

4 Results

4.1 Which factors contribute to polarization?

Table 2 Aposteriori unimodality results for the dices-350 dataset.

SDB Feature	Value	apunim	support
Age	1) Gen. X+	0.0521***	10819
	2) Millenial	-0.0090	12564
	3) Gen. Z	-0.0259	19544
Education	College degree or higher	0.0308	26175
	High school or below	-0.0581***	14309
	Other	-0.0006	2443
Gender	Man	0.0037	21289
	Woman	-0.0054	21638
Race	African American	-0.1022***	10121
	Asian	0.0661***	9074
	Latino	0.0344*	7678
	Multiracial	0.0025	5584
	White	-0.0028	10470

Table 3 Aposteriori unimodality results for the dices-990 dataset.

SDB Feature	Value	apunim	support
Age	1) Gen. X+	-0.0639***	15113
	2) Millenial	-0.0133	40187
	3) Gen. Z	0.1076***	14396
Education	College degree or higher	0.0144	61925
	High school or below	-0.0843***	7771
Gender	Man	0.0222*	34149
	Woman	-0.0157	34920
Race	African American	0.1093***	6096
	Asian	0.0555***	19258
	Latino	-0.1962***	6935
	Other	0.0673***	25494
	White	-0.1480***	11913

Table 4 Aposteriori unimodality results for the sap dataset.

SDB Feature	Value	apunim	support
Age	1) Gen. X+	0.0192	743
	2) Millennial	0.0189	1792
	3) Gen. Z	-0.0153	239
Ethnicity	black	-0.2967***	1041
	white	0.1400***	2048
Gender	man	0.0942***	1635
	woman	-0.1810***	1333

Table 5: Aposteriori unimodality results for the kumar dataset.

SDB Feature	Value	apunim	support
Age	1) 18 - 24	-0.0090	9742
	2) 25 - 34	0.0324***	33203
	3) 35 - 44	-0.0093	21055
	4) 45 - 54	0.0006	11025
	5) 55 - 64	-0.0037	6123
	6) 65 or older	0.0001	2567
Education	1) Doctoral degree	0.0048	1019
	2) Professional degree	0.0038	1321
	3) Master's degree	0.0147**	13420
	4) Bachelor's degree	0.0401***	34085

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Table 5: Aposteriori unimodality results for the kumar dataset.

SDB Feature	Value	apunim	support
	5) Associate degree	-0.0113**	9122
	6) College, no degree	-0.0335***	16666
	7) High School graduate	-0.0079*	7326
	8) No high school	0.0004	462
Ethnicity	Asian	-0.0019	5062
	Black	0.0110*	10999
	Hispanic	-0.0025	2389
	Multiracial	-0.0041	3682
	Other	0.0022	1507
	Unknown	-0.0030	1160
	White	-0.0377***	45306
Gender	Female	-0.0278***	40686
	Male	0.0275***	37929
	Nonbinary	-0.0008	489
	Unknown	-0.0029	621
Has Been Targeted	False	-0.0633***	45234
	True	0.0349***	24491
Is Parent	No	-0.0707***	37231
	Prefer not to say	-0.0023	831
	Yes	0.0535***	41493
Is Transgender	No	-0.0894***	11978
	Prefer not to say	0.0005	856
	Yes	0.0013	2306
Political Affiliation	Conservative	0.0331***	23010
	Independent	-0.0157***	22818
	Liberal	-0.0287***	32917
	Other	-0.0049	1822
	Prefer not to say	0.0015	3548
Religion Important	1) No	-0.0804***	26008
	2) Not very	-0.0154***	10055
	3) Somewhat	0.0210***	20169
	4) Very	0.0487***	27284
	Prefer not to say	0.0001	1179
Seen Toxicity	False	0.0311***	20185
	True	-0.0717***	43265
Sexual Orientation	Bisexual	0.0175***	10005
	Heterosexual	-0.0571***	38048
	Homosexual	-0.0017	2975
	Other	-0.0003	728

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Table 5: Aposteriori unimodality results for the kumar dataset.

SDB Feature	Value	apunim	support
	Prefer not to say	-0.0017	1751
	NaN	0.0031	208
Technology Impact	1) Very negative	-0.0009	784
	2) Somewhat negative	-0.0080	7973
	3) Neutral	0.0027	10720
	4) Somewhat positive	-0.0265***	40477
	5) Very positive	0.0208***	22471
Toxicity Problem	1) Never	0.0058	4962
	2) Rarely	0.0177***	14457
	3) Occasionally	-0.0058	29476
	4) Frequently	-0.0217***	24774
	5) Very Frequently	-0.0145***	10656

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided SDB dimensions can partially explain polarization in the toxicity/racism annotations. The results for the Kumar and Sap datasets can be found in Tables 4 and 5 respectively, while Tables 2 and 3 display the results for the DICES-350 and DICES-990 datasets respectively.

We observe statistically significant polarization effects caused by annotator* Ethnicity *on all datasets

In the DICES dataset, non-white annotators seem to contribute to polarization much more than the majority white annotators. This effect is reversed for the [4] dataset.

Annotator* age *does not cause polarization for toxicity/hate detection, but does for safety

None of the age groups in the Sap and Kumar datasets produce statistically significant results for polarization attribution, suggesting that it is not an important factor. However, both DICES datasets produced statistically and quantitatively significant results. This may be caused by the large number of annotators in these datasets, which enable our method to better distinguish this effect from noise in the annotations. This is unlikely however, given our next finding.

Polarization attribution follows ordinal ranking

While most SDB attributes are categorical, some are ordinal (e.g., Age, Education, LIKERT-scale opinions). We discover a pattern among (statistically significant) ordinal attributes, where polarization changes monotonically with their ranking. Examples include:

- Polarization increasing/decreasing with annotator age in both DICES datasets (but not in the Sap and Kumar datasets, where there is no statistical significance).

- Thoughts on religion and toxicity in the Kumar dataset (although much more noisily, since some attribute factors do not display statistical significance).

A notable exception is *Education* in the Kumar dataset. This could be attributed to the curse of dimensionality, since we investigate a great number of groups, but there are few annotators for each individual group.

The fact that this behavior is not replicated when investigating annotator age in the Sap and Kumar datasets lends further support to the argument that age does indeed not play a major role in annotator polarization for these datasets.

4.2 Investigating ordinal attributes

Analysis on annotated data frequently ignores the different kinds of measurements in studies using sociodemographic and LIKERT-style attributes. Researchers often treat all dimensions as *nominal/categorical* even though some are inherently ordinal (e.g., LIKERT, education), and some continuous/ratio variables (e.g., age). By treating these variables as categorical, not only do we lose valuable information, but we may magnify statistical errors [16].

Some researchers attempt to find an "optimal point" which can be used to binarize continuous variables [**?**], but this approach is misguided. By using any artificial cut-off point, the resulting analysis is immediately invalidated by any change in the underlying data. Furthermore, some of these approaches tend to produce ouroboroid[3] results; a statistical measure is used to binarize the data during preprocessing, which is then fed to a methodology utilizing the same measure to get results. This is methodologically unsound, since it biases the model by transforming the data in a way that model best recognizes to infer a specific outcome. In this work, we argue that *retaining the order of ordinal attributes can reveal interesting trends* that would be otherwise missed with categorical inference.

Most ordinal attributes (with statistically significant apunim values) tend to cluster around a few dominant values; for example, *Age* and *Education* in the Kumar dataset. This pattern is expected because the Apunim algorithm is designed to identify clusters of annotations, causing nearby or similar values to be absorbed into the main effects driven by a few significant modes.

However, some ordinal attributes (*Age* in both DICES-350 and DICES-990, *Religion Importance* in the Kumar dataset follow a strict, monotonic trend. This implies that, in some cases, the **amount of polarization increases the further apart an opinion is from neutrality**. For example, non-religious annotators are less likely to have polarizing opinions about what constitutes a toxic comment than the average annotator, while very religious annotators generate very polarizing judgments.

5 How many annotators?

While our test works best with many annotations per datapoint, the cost required for multiple human annotators is often prohibitive [11]. It is thus very important both DICES datasets, as well as the Sap dataset annotated by 100 distinct LLM

[3] "Snake eating its own tail".

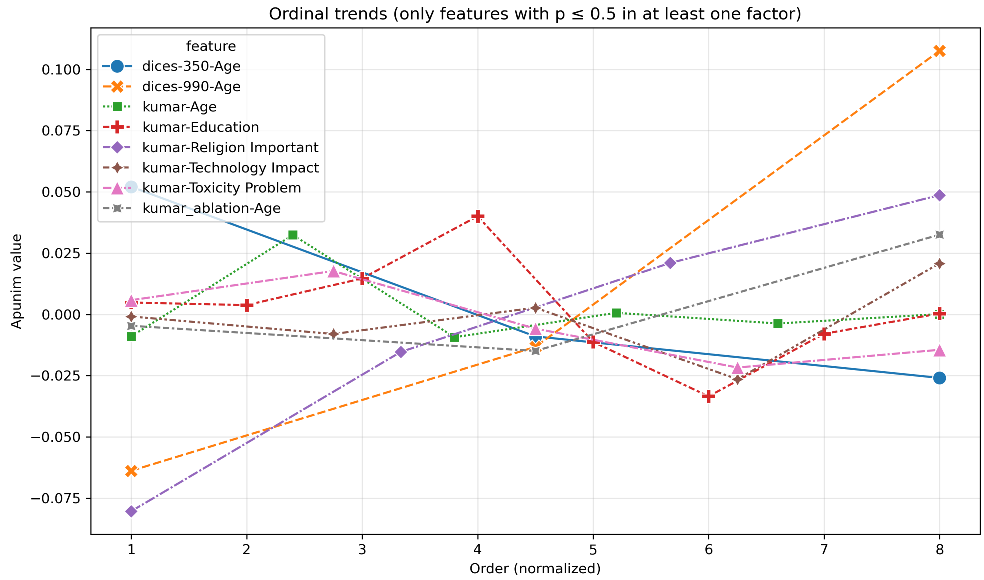

Fig. 7 The behavior of ordinal variables in all datasets. The x-axis represents the order of each factor (marked as `1), 2), ...` in Tables 2,3,5,4). The x-axis is normalized to the smallest common denominator between the number of ordinal values.

annotators. While a more appropriate comparison would entail synthetic annotations in the DICES datasets themselves, we elect to use a standard task for LLM-as-a-judge (hate speech detection) instead of the relatively challenging and under-explored area of general LLM safety.

For each comment in all datasets, we randomly sample a subset of annotators and compute the apunim value. We repeat the procedure 30 times, and average all observations. This was a necessary step to reduce the variance inherent in random sampling. We also truncate the results for the DICES-350 dataset, since very few comments were annotated by the respective annotator counts.

Figure 8 demonstrates the effect of the number of annotators on the reliability of the polarization estimation. We can see that the metric becomes consistent with about 20 annotators per comment, after which we start to get diminishing results. This experiment was conducted on a subset of the Sap dataset, with LLM annotators having different SDBs.

6 Conclusion

We introduced the "Aposteriori Unimodality Test", a statistical test that detects whether certain sociodemographic groups partly cause polarization in annotations. Our test is resistant to low samples of annotations, bypasses common statistical issues such as sample dependence, and avoids newly-discovered issues such as the presence of inherent polarization and polarization direction. We apply our test two

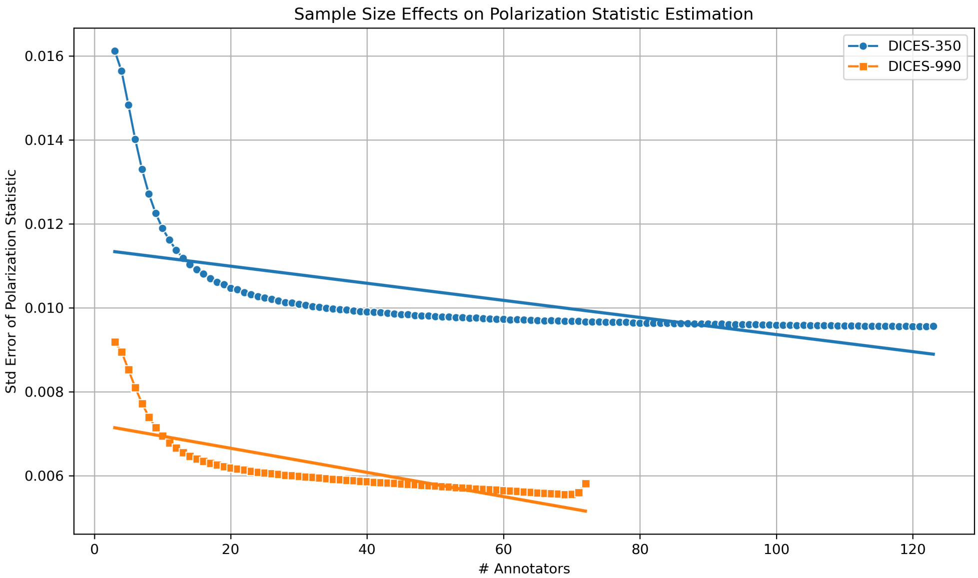

Fig. 8 Standard error of the *pol-statistic* when sampled with replacement for various number of annotators.

human-generated and two LLM-generated datasets and find interesting patterns on the former, and verify current literature on LLM SDB prompting on the latter.

7 Limitations

Since there are no other established tests that attribute polarization to sociodemographic characteristics, there is no way to verify that the results presented in §4 are accurate. Similarly, there is no qualitative way of evaluating our test, other than observations on the (non-) efficacy of LLM SDB prompting and our own intuition. Lastly, while our test seems to work sufficiently well with a small number of annotators per data-point, we still encourage researchers using this test to have at least a few annotators of each sociodemographic group that is being tested, in their dataset.

8 Ethical Considerations

While our work aims to help protect marginalized and disadvantaged groups by attributing polarization to certain subgroups, it can be taken advantage of by malicious actors. Given a dataset with fine-grained SDB information, these actors can use our test to target specific vulnerable groups. We thus urge researchers to keep datasets including such information protected, and provided to others only under explicit permission.

Appendix A Acronyms

LLM Large Language Model
SDB SocioDemographic Background
nDFU normalized Distance From Unimodality
FWER Family-Wise Error Rate

Appendix B Theoretical limitations

In §2.2 we compared the polarization of the whole set of annotations with the partitions by annotator group. This approach uses the same samples for both subset and superset, violating the fundamental hypothesis of independent samples. This also means the statistics for both will *always be similar to an extent.* This is especially pronounced in our case, due to the extreme overlap exhibited by tiny annotation sets. Additionally, when calculating p-values, the non-independence of samples induces an intrinsic, non-zero covariance between any statistic s calculated on A and on S.

A standard method of side-stepping this issue is to compare the subset with its complement to the superset. This can not be applied to our case. Consider the example in Figure 3; if we compared $nDFU_{men}$ with $nDFU_{women}$, the result would be very close to zero, indicating no polarization attribution to gender.

Appendix C How do we handle continuous variables?

Continuous attributes (such as age) can be discretized during analysis; this practice should generally be avoided [16, 17], although it is necessary when the data has extremely skewed distributions [16], the underlying data are already in discretized form, or when the tools used for analysis require some form of categorization. Our methodology is built upon nDFU, and thus requires histograms to be well-defined. Additionally, the data for Age is already discretized for the DICES and Kumar datasets; for the sake of comparison, we discretize the Sap dataset with the same method as the DICES dataset, instead of using generalized binning techniques based on statistics [18–20].

In order to make sure our results are not biased, we conduct ablation experiments with all ordinal variables from the Kumar dataset discretized to three factors. We observe that...

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Table C1 Aposteriori unimodality results for the kumar_ablation dataset.

SDB Feature	Value	apunim	support
Age	1) Gen. X+	-0.0047	8690
	2) Millennial	-0.0149**	31470
	3) Gen. Z	0.0325***	40951
Education	Neutral	0.0150***	13919
	No	0.0000	2340
	Yes	-0.0376***	39926
Religion Important	Neutral	0.0226***	21348
	No	-0.1111***	35028
	Yes	0.0485***	27284
Technology Impact	Neutral	0.0026	10720
	No	-0.0086*	8757
	Yes	0.0114*	42443
Toxicity Problem	Neutral	-0.0060	29476
	No	0.0255***	19334
	Yes	-0.0399***	34645

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